Brutal Series - Paper B63 (Respiration, Decimals, Weather & Climate, Exponents)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A boy breathes 18 times a minute at rest. During brisk exercise his breathing rate becomes two and a half times the resting rate. How many MORE breaths does he take in 3 minutes of exercise than he would have at rest in the same time? [Respiration in Organisms]

- **(A)** 81
- **(B)** 45
- **(C)** 108
- **(D)** 162
- **(E)** 135
- **(F)** 27
- **(G)** 90

2. A ribbon is 12.5 metres long. A tailor cuts pieces each 0.75 metre long for collars. After cutting as many whole pieces as possible, how much ribbon is left over? [A Peek Beyond the Point]

- **(A)** 1.25 m
- **(B)** 0.25 m
- **(C)** 0.5 m
- **(D)** 16 pieces
- **(E)** 0.75 m
- **(F)** 0 m

3. A weather station records these maximum temperatures (in degrees Celsius) over 5 days: 31, 29, 34, 30, 26. What is the average maximum temperature for these days? [Weather, Climate & Adaptations]

- **(A)** 150 deg C
- **(B)** 26 deg C
- **(C)** 30 deg C
- **(D)** 29 deg C
- **(E)** 34 deg C
- **(F)** 8 deg C

4. Simplify (2^3 x 2^4) / 2^5 and write the answer as a single number. [Exponents & Powers]

- **(A)** 2
- **(B)** 64
- **(C)** 32
- **(D)** 2^2 only
- **(E)** 8
- **(F)** 4
- **(G)** 16

5. During one breath a person draws in 500 mL of air. Inhaled air is 21% oxygen but exhaled air is only 16% oxygen. How many millilitres of oxygen are absorbed by the body from one 500 mL breath? [Respiration in Organisms]

- **(A)** 25 mL
- **(B)** 20 mL
- **(C)** 105 mL
- **(D)** 5 mL
- **(E)** 185 mL
- **(F)** 80 mL

6. A bottle holds 1.5 litres of juice. Glasses of 0.125 litre are filled from it. How many full glasses can be poured, and how much juice is left? [A Peek Beyond the Point]

- **(A)** 12 glasses, 0 litre left
- **(B)** 8 glasses, 0.5 litre left
- **(C)** 12.5 glasses
- **(D)** 12 glasses, 0.5 litre left
- **(E)** 11 glasses, 0.125 litre left
- **(F)** 15 glasses, 0 litre left
- **(G)** 10 glasses, 0.25 litre left

7. On one day the maximum temperature is 34 degrees Celsius and the minimum is 19 degrees Celsius. The next day the maximum rises by 3 and the minimum falls by 4. What is the temperature RANGE (max minus min) on the second day? [Weather, Climate & Adaptations]

- **(A)** 8 deg C
- **(B)** 15 deg C
- **(C)** 14 deg C
- **(D)** 37 deg C
- **(E)** 20 deg C
- **(F)** 16 deg C
- **(G)** 22 deg C

8. Write 0.00056 in standard form (a x 10^n where 1 <= a < 10). [Exponents & Powers]

- **(A)** 0.56 x 10^-3
- **(B)** 5.6 x 10^-4
- **(C)** 5.6 x 10^-5
- **(D)** 56 x 10^-5
- **(E)** 5.6 x 10^4
- **(F)** 560 x 10^-6

9. In yeast, sugar breaks down without oxygen to give alcohol and a gas. A baker finds that each gram of sugar releases 0.5 litre of this gas. If the dough is given 24 grams of sugar and only three-quarters of it is used up, how many litres of gas are produced? [Respiration in Organisms]

- **(A)** 9.5 litres
- **(B)** 6 litres
- **(C)** 12 litres
- **(D)** 3 litres
- **(E)** 36 litres
- **(F)** 9 litres

10. Find the value of 4.2 x 0.05 + 3.6 / 0.9 (do multiplication and division before adding). [A Peek Beyond the Point]

(A) 8.0
(B) 4.0
(C) 4.21
(D) 0.42
(E) 0.421
(F) 4.6

11. Climate is the average weather over many years. A town's rainfall over 3 months is 45 mm, 60 mm and 75 mm. If this pattern is the same every year, what is the average MONTHLY rainfall? [Weather, Climate & Adaptations]

(A) 20 mm
(B) 75 mm
(C) 180 mm
(D) 30 mm
(E) 45 mm
(F) 60 mm
(G) 90 mm

12. Find the value of 3^4 - 4^3. [Exponents & Powers]

(A) 1
(B) -17
(C) 0
(D) 145
(E) 17
(F) 81

13. A runner's muscles run short of oxygen and switch to anaerobic respiration, making lactic acid that causes cramps. If aerobic respiration of one glucose unit gives 38 energy units but anaerobic gives only 2, how many TIMES more energy does the aerobic path give from the same glucose? [Respiration in Organisms]

(A) 20 times
(B) 18 times
(C) 2 times
(D) 40 times
(E) 38 times
(F) 19 times

14. A car uses 0.08 litre of petrol per kilometre. Petrol costs Rs 102.5 per litre. What is the fuel cost for a 25 km trip? [A Peek Beyond the Point]

(A) Rs 205
(B) Rs 410
(C) Rs 164
(D) Rs 102.5
(E) Rs 20.5
(F) Rs 256.25
(G) Rs 2050

15. Polar bears have a thick fat layer and dense fur to survive the cold. If a polar region's temperature is -25 degrees Celsius and a tropical region's is 35 degrees Celsius, by how many degrees is the polar region colder? [Weather, Climate & Adaptations]

- **(A)** 25 deg C
- **(B)** -60 deg C
- **(C)** 35 deg C
- **(D)** 60 deg C
- **(E)** 10 deg C
- **(F)** 50 deg C
- **(G)** 70 deg C

16. Express the product 100000 x 1000 as a power of 10, then state how many zeros the answer has. [Exponents & Powers]

- **(A)** 10^15, with 15 zeros
- **(B)** 10^9, with 9 zeros
- **(C)** 10^6, with 6 zeros
- **(D)** 10^5, with 5 zeros
- **(E)** 10^7, with 7 zeros
- **(F)** 10^8, with 7 zeros
- **(G)** 10^8, with 8 zeros

17. A fish takes in water through its mouth and pushes it over the gills. In an experiment a fish passes 240 mL of water over its gills each minute and absorbs 8% of the oxygen dissolved in it. If each 240 mL holds 6 mL of dissolved oxygen, how much oxygen does the fish absorb in 5 minutes? [Respiration in Organisms]

- **(A)** 2.88 mL
- **(B)** 2.4 mL
- **(C)** 1.2 mL
- **(D)** 4.8 mL
- **(E)** 24 mL
- **(F)** 0.48 mL

18. The product 0.6 x 0.6 x 0.6 is calculated. By how much is the answer LESS than 0.6? [A Peek Beyond the Point]

- **(A)** 0.216
- **(B)** 1.8
- **(C)** 0.18
- **(D)** 0.416
- **(E)** 0.6
- **(F)** 0.384

19. A humidity sensor shows the air holds 18 grams of water vapour per cubic metre, but at that temperature it could hold 24 grams at most. What is the relative humidity as a percentage? [Weather, Climate & Adaptations]

- **(A)** 18%
- **(B)** 42%
- **(C)** 24%
- **(D)** 60%
- **(E)** 75%
- **(F)** 6%

20. Simplify (5^2)^3 / 5^4 as a power of 5, then give its value. [Exponents & Powers]

- **(A)** 5^2 = 25
- **(B)** 5^2 = 10
- **(C)** 5^6 = 15625
- **(D)** 5^1 = 5
- **(E)** 5^10
- **(F)** 5^4 = 625

21. Cockroaches breathe through tiny holes called spiracles along the body, leading into air tubes. A cockroach has 10 pairs of spiracles. If a scientist seals 30% of all its spiracles with wax, how many spiracles are left open? [Respiration in Organisms]

- **(A)** 14
- **(B)** 13
- **(C)** 6
- **(D)** 7
- **(E)** 10
- **(F)** 3
- **(G)** 16

22. Three parcels weigh 2.45 kg, 1.8 kg and 0.755 kg. They are packed into a box that itself weighs 0.495 kg. What is the total weight to be carried? [A Peek Beyond the Point]

- **(A)** 5.0 kg
- **(B)** 5.05 kg
- **(C)** 6.5 kg
- **(D)** 5.55 kg
- **(E)** 4.95 kg
- **(F)** 5.5 kg

23. The maximum temperature is 28 degrees Celsius and the minimum is 16 degrees Celsius. The 'mean' daily temperature is taken as the average of the maximum and minimum. What is it? [Weather, Climate & Adaptations]

- **(A)** 16 deg C
- **(B)** 12 deg C
- **(C)** 22 deg C
- **(D)** 20 deg C
- **(E)** 24 deg C
- **(F)** 28 deg C
- **(G)** 44 deg C

24. Each time a thin cloth is folded over once, its number of layers doubles. Starting from a single layer, how many layers are stacked up after 7 such folds? [Exponents & Powers]

- **(A)** 14
- **(B)** 256
- **(C)** 64
- **(D)** 127
- **(E)** 128
- **(F)** 2
- **(G)** 49

25. Lime water turns milky when carbon dioxide is bubbled through it. A student breathes out into lime water and it goes milky in 8 seconds. After light jogging the same volume of breath turns it milky in only 2 seconds. How many times faster did the milkiness appear after jogging? [Respiration in Organisms]

- **(A)** 0.25 times
- **(B)** 10 times
- **(C)** 6 times
- **(D)** 4 times
- **(E)** 16 times
- **(F)** 8 times

26. A number becomes 73.5 when multiplied by 100 and then 0.15 is added. What was the original number? [A Peek Beyond the Point]

- **(A)** 0.07335
- **(B)** 0.735
- **(C)** 7.335
- **(D)** 73.35
- **(E)** 0.7485
- **(F)** 7.35
- **(G)** 0.7335

27. Tropical rainforests get heavy rain. A rain gauge collects 8 mm in the first hour and the rate then doubles each hour. How much rain falls during the THIRD hour alone? [Weather, Climate & Adaptations]

- **(A)** 16 mm
- **(B)** 48 mm
- **(C)** 8 mm
- **(D)** 32 mm
- **(E)** 64 mm
- **(F)** 56 mm
- **(G)** 24 mm

28. Work out and add together these two powers of negative two: (-2)^4 plus (-2)^3. What single number do you get? [Exponents & Powers]

- **(A)** 8
- **(B)** 24
- **(C)** -24
- **(D)** 16
- **(E)** -8
- **(F)** 0

29. Plants respire all the time but in daylight photosynthesis hides it. A leaf releases 12 mg of carbon dioxide per hour by respiration, while in light it absorbs 30 mg per hour for photosynthesis. What is the NET carbon dioxide exchange of the leaf per hour in daylight? [Respiration in Organisms]

- **(A)** 18 mg released net
- **(B)** 2.5 mg absorbed
- **(C)** 18 mg released
- **(D)** 18 mg absorbed
- **(E)** 30 mg absorbed
- **(F)** 42 mg absorbed

30. A wall is 3.2 metres long. A photo of it is drawn so that 1 metre on the wall is shown as 0.025 metre on paper. How long is the wall in the drawing, in centimetres (1 m = 100 cm)? [A Peek Beyond the Point]

- **(A)** 0.08 cm
- **(B)** 80 cm
- **(C)** 8 cm
- **(D)** 12.8 cm
- **(E)** 0.8 cm
- **(F)** 16 cm

31. Animals of the tropical rainforest are adapted to climb and live on trees. In a survey of 200 animals, 35% live on the forest floor and the rest live on trees. How many MORE tree-dwellers are there than floor-dwellers? [Weather, Climate & Adaptations]

- **(A)** 100
- **(B)** 70
- **(C)** 130
- **(D)** 60
- **(E)** 65
- **(F)** 30

32. The number 3,40,00,000 (three crore forty lakh) is written in standard form as 3.4 x 10^n. What is n? [Exponents & Powers]

- **(A)** 9
- **(B)** 7
- **(C)** 6
- **(D)** 5
- **(E)** 10
- **(F)** 4
- **(G)** 8

33. Breathing rate rises with body size differences in activity. A girl breathes 20 times per minute. Each breath moves 0.4 litre of air. How many litres of air does she breathe in one quarter of an hour? [Respiration in Organisms]

- **(A)** 48 litres
- **(B)** 300 litres
- **(C)** 60 litres
- **(D)** 120 litres
- **(E)** 480 litres
- **(F)** 12 litres
- **(G)** 8 litres

34. Evaluate 9 - 0.45 / 0.05 (division first). [A Peek Beyond the Point]

- **(A)** 0
- **(B)** 0.45
- **(C)** 18
- **(D)** -0.9
- **(E)** 8.55
- **(F)** 0.9
- **(G)** 9

35. A migratory bird flies 1500 km to escape cold winters. It covers 250 km each day but rests one day after every 2 flying days. How many days does the whole journey take? [Weather, Climate & Adaptations]

- **(A)** 12 days
- **(B)** 7 days
- **(C)** 5 days
- **(D)** 9 days
- **(E)** 8 days
- **(F)** 10 days

36. Simplify 2^0 + 3^0 + (5 x 7)^0 + 4^1. [Exponents & Powers]

- **(A)** 4
- **(B)** 3
- **(C)** 8
- **(D)** 7
- **(E)** 5
- **(F)** 0
- **(G)** 10

37. The diaphragm is a muscle that helps in breathing. When we breathe in, it moves down and flattens, increasing chest space. In 1 minute a resting person makes 16 such complete breaths. If each breath in plus breath out takes the same total time, how many seconds does ONE full breath last? [Respiration in Organisms]

- **(A)** 3.75 seconds
- **(B)** 16 seconds
- **(C)** 0.27 seconds
- **(D)** 7.5 seconds
- **(E)** 1.875 seconds
- **(F)** 3.6 seconds

38. A runner covers 0.4 km in 1 minute. A cyclist covers 0.65 km in the same minute. In 12 minutes, how much FARTHER does the cyclist travel than the runner? [A Peek Beyond the Point]

- **(A)** 7.8 km
- **(B)** 4.8 km
- **(C)** 12.6 km
- **(D)** 2.5 km
- **(E)** 0.25 km
- **(F)** 0.3 km
- **(G)** 3 km

39. A weather report says there is a 0.2 chance of rain on each of 3 separate, unrelated days. Out of 100 such 3-day stretches, how many would you expect to have NO rain on day 1 (using the day-1 chance only)? [Weather, Climate & Adaptations]

- **(A)** 80
- **(B)** 100
- **(C)** 2
- **(D)** 60
- **(E)** 20
- **(F)** 8

40. Which is greater and by how much: 2^6 or 6^2? [Exponents & Powers]

- **(A)** 6^2 is greater by 28
- **(B)** 2^6 is greater by 36
- **(C)** 2^6 is greater by 28
- **(D)** 6^2 is greater by 12
- **(E)** 2^6 is greater by 12
- **(F)** they are equal
- **(G)** 2^6 is greater by 64

41. An earthworm has no lungs and breathes through its moist skin. If 0.5 mg of oxygen passes through each square centimetre of skin per hour, and a worm has 36 square centimetres of breathing skin, how much oxygen does it take in over 2 hours? [Respiration in Organisms]

- **(A)** 9 mg
- **(B)** 32 mg
- **(C)** 144 mg
- **(D)** 72 mg
- **(E)** 18 mg
- **(F)** 36 mg
- **(G)** 12 mg

42. A shopkeeper sells cloth at Rs 56.25 per metre. A customer buys 3.6 metres and pays with a Rs 250 note. How much change is returned? [A Peek Beyond the Point]

- **(A)** Rs 56.25
- **(B)** Rs 250
- **(C)** Rs 47.5
- **(D)** Rs 47.5 short
- **(E)** Rs 52.5
- **(F)** Rs 202.5

43. Desert animals stay underground in the day to avoid heat. If the desert surface is 47 degrees Celsius but a burrow 1 metre down is 24 degrees Celsius, and the temperature drops steadily with depth, how much cooler is the burrow than the surface? [Weather, Climate & Adaptations]

- **(A)** 71 deg C
- **(B)** 24 deg C
- **(C)** 47 deg C
- **(D)** 23 deg C
- **(E)** 13 deg C
- **(F)** 2 deg C

44. A bacterium splits into 2 every hour. Starting with 5 bacteria, how many are there after 4 hours? [Exponents & Powers]

- **(A)** 160
- **(B)** 16
- **(C)** 45
- **(D)** 40
- **(E)** 20
- **(F)** 80
- **(G)** 625

45. Glucose is the fuel of respiration. Suppose burning one glucose unit needs 6 units of oxygen and gives out 6 units of carbon dioxide. A cell completely respires 15 glucose units. How many units of oxygen and carbon dioxide are involved IN TOTAL (oxygen used plus carbon dioxide made)? [Respiration in Organisms]

- **(A)** 160 units
- **(B)** 90 units
- **(C)** 180 units
- **(D)** 360 units
- **(E)** 12 units
- **(F)** 45 units
- **(G)** 36 units

46. Which is the largest: 0.7, 0.07, 0.707, 0.077, 0.7007? After choosing it, subtract the SMALLEST of the five from it. [A Peek Beyond the Point]

- **(A)** 0.637
- **(B)** 0.077
- **(C)** 0.7007
- **(D)** 0.7
- **(E)** 0.63
- **(F)** 0.6363
- **(G)** 0.6437

47. Over a week the rainfall (in mm) is 12, 0, 8, 20, 0, 5, 10. On how many days was the rainfall ABOVE the weekly average? [Weather, Climate & Adaptations]

- **(A)** 6 days
- **(B)** 7 days
- **(C)** 1 day
- **(D)** 3 days
- **(E)** 5 days
- **(F)** 4 days
- **(G)** 2 days

48. Express 72 as a product of prime powers. [Exponents & Powers]

- **(A)** 2^4 x 3
- **(B)** 2^3 x 3^2
- **(C)** 2^3 x 3^3
- **(D)** 2^2 x 3^3
- **(E)** 2 x 3^3
- **(F)** 6^2 x 2

49. Roots also respire and need air in the soil. A potted plant's roots use 0.25 litre of oxygen a day. After heavy watering, the waterlogged soil cuts the oxygen supply by 60%. How many litres of oxygen can the roots get on that day? [Respiration in Organisms]

- **(A)** 0.2 litre
- **(B)** 0.25 litre
- **(C)** 0.06 litre
- **(D)** 0.4 litre
- **(E)** 0.15 litre
- **(F)** 0.1 litre

50. A medicine dose is 0.005 litre. A full bottle holds 0.6 litre. After 80 doses are given, how much medicine is left in the bottle? [A Peek Beyond the Point]

- **(A)** 0.16 litre
- **(B)** 0.04 litre
- **(C)** 0.595 litre
- **(D)** 0.2 litre
- **(E)** 0.4 litre
- **(F)** 0.5 litre

51. A camel can drink 80 litres of water in one go. It loses water at 4 litres per day in the desert. If it starts full after drinking, for how many days can it last before the 80 litres run out? [Weather, Climate & Adaptations]

- **(A)** 20 days
- **(B)** 25 days
- **(C)** 16 days
- **(D)** 320 days
- **(E)** 4 days
- **(F)** 40 days

52. Reduce the expression (3^5 x 3^2) divided by (3^3 x 3^2) to a single power of three, and then state its numerical value. [Exponents & Powers]

- **(A)** 81
- **(B)** 1
- **(C)** 9
- **(D)** 27
- **(E)** 6
- **(F)** 3
- **(G)** 243

53. After a sprint a person breathes fast to repay an 'oxygen debt'. A runner's normal breathing is 16 per minute. For 4 minutes after the race it stays raised at 40 per minute, then returns to normal. How many EXTRA breaths (above normal) were taken during those 4 minutes? [Respiration in Organisms]

- **(A)** 100
- **(B)** 160
- **(C)** 64
- **(D)** 96
- **(E)** 32
- **(F)** 24
- **(G)** 144

54. Compute (2.5 + 1.5) x (0.6 - 0.35). Use brackets first. [A Peek Beyond the Point]

- **(A)** 4
- **(B)** 2.25
- **(C)** 10
- **(D)** 16
- **(E)** 0.25
- **(F)** 0.1
- **(G)** 1

55. A place has a hot climate if its yearly average temperature is above 25 degrees Celsius. Four months read 22, 28, 30 and 24 degrees Celsius. Assuming these repeat all year, is the climate hot, and what is the average? [Weather, Climate & Adaptations]

- **(A)** Average 25 deg C, so hot
- **(B)** Average 26 deg C, exactly the limit
- **(C)** Average 30 deg C, so hot
- **(D)** Average 26 deg C, not hot
- **(E)** Average 26 deg C, so hot
- **(F)** Average 24 deg C, not hot
- **(G)** Average 104 deg C, so hot

56. Light from the Sun travels about 1.5 x 10^8 km to Earth. Write this distance as an ordinary number (how many km). [Exponents & Powers]

- **(A)** 15000000 km
- **(B)** 150000000 km
- **(C)** 108000000 km
- **(D)** 1500000 km
- **(E)** 1500000000 km
- **(F)** 15 x 10^7 km only
- **(G)** 150000 km

57. Gills give fish a huge surface to absorb oxygen. Suppose a fish's gills have a surface of 4 times its skin area, and the skin area is 30 square centimetres. If oxygen is absorbed at 0.2 mg per square centimetre of gill per minute, how much oxygen do the gills absorb in 1 minute? [Respiration in Organisms]

- **(A)** 120 mg
- **(B)** 6 mg
- **(C)** 240 mg
- **(D)** 12 mg
- **(E)** 48 mg
- **(F)** 2.4 mg
- **(G)** 24 mg

58. A water tank leaks 0.125 litre every minute. How many litres leak away in 1 hour and 20 minutes? [A Peek Beyond the Point]

- **(A)** 12 litres
- **(B)** 16 litres
- **(C)** 10 litres
- **(D)** 2.5 litres
- **(E)** 7.5 litres
- **(F)** 8 litres
- **(G)** 1.0 litre

59. Penguins huddle to keep warm. A single penguin loses 30 units of heat per minute, but huddling cuts each one's loss by 40%. How much heat does one penguin lose in 10 minutes while huddling? [Weather, Climate & Adaptations]

- **(A)** 420 units
- **(B)** 210 units
- **(C)** 300 units
- **(D)** 108 units
- **(E)** 18 units
- **(F)** 180 units
- **(G)** 120 units

60. Find the value of 10^4 / 10^2 + 10^1. [Exponents & Powers]

- **(A)** 110
- **(B)** 20
- **(C)** 10010
- **(D)** 11
- **(E)** 100
- **(F)** 1010

61. Exhaled air carries water vapour, seen as mist on a cold mirror. A person loses 0.6 mL of water per minute as vapour while breathing. During 90 minutes of sleep, how many millilitres of water are lost through breathing? [Respiration in Organisms]

- **(A)** 54 mL
- **(B)** 60 mL
- **(C)** 15 mL
- **(D)** 45 mL
- **(E)** 90 mL
- **(F)** 5.4 mL

62. A decimal number is rounded to 2.7 when written to one decimal place. Which of these could be the original number: 2.74, 2.75, 2.66, 2.649, 2.751? [A Peek Beyond the Point]

- **(A)** 2.649
- **(B)** 2.751
- **(C)** 2.7
- **(D)** 2.66
- **(E)** 2.8
- **(F)** 2.74
- **(G)** 2.75

63. A barometer measures air pressure. As a storm approaches, pressure drops from 1012 units by 3 units each hour. After how many hours will it reach 991 units? [Weather, Climate & Adaptations]

- **(A)** 7 hours
- **(B)** 3 hours
- **(C)** 10 hours
- **(D)** 9 hours
- **(E)** 6 hours
- **(F)** 21 hours
- **(G)** 8 hours

64. A square has side 2^3 cm. What is its area written as a power of 2 (in square cm), and its value? [Exponents & Powers]

- **(A)** 2^9 = 512
- **(B)** 2^5 = 32
- **(C)** 2^3 = 8
- **(D)** 2^6 = 36
- **(E)** 2^4 = 16
- **(F)** 2^6 = 64
- **(G)** 4^3 = 64 as 4 power

65. Muscle cells store energy released by respiration. If respiring 1 gram of glucose releases enough energy to lift a 4 kg load 100 metres, how high could the same energy lift a 4 kg load if instead it were spread to lift it many small times totalling 0.8 gram of glucose worth? [Respiration in Organisms]

- **(A)** 100 metres
- **(B)** 125 metres
- **(C)** 80 metres
- **(D)** 8 metres
- **(E)** 20 metres
- **(F)** 320 metres
- **(G)** 64 metres

66. A rod is cut into pieces of 0.6 m, 0.45 m and 0.95 m, and 0.1 m is lost as sawdust at each of the 2 cuts. What was the original length of the rod? [A Peek Beyond the Point]

- **(A)** 2.4 m
- **(B)** 2.05 m
- **(C)** 2.1 m
- **(D)** 2.2 m
- **(E)** 2.3 m
- **(F)** 2.0 m
- **(G)** 1.8 m

67. Tropical fish are adapted to warm water. A tank must stay between 24 and 28 degrees Celsius. The water is at 21 degrees Celsius and a heater raises it by 0.5 degree every 6 minutes. How long until it FIRST reaches the safe range (24 degrees)? [Weather, Climate & Adaptations]

- **(A)** 18 minutes
- **(B)** 6 minutes
- **(C)** 42 minutes
- **(D)** 30 minutes
- **(E)** 36 minutes
- **(F)** 48 minutes

68. Simplify (2^2 x 3^2) and compare it with 6^2. What do you find? [Exponents & Powers]

- **(A)** 2^2 x 3^2 = 13, 6^2 = 36
- **(B)** both equal 12
- **(C)** both equal 72
- **(D)** 2^2 x 3^2 = 36, 6^2 = 12
- **(E)** both equal 6^4
- **(F)** Both equal 36

69. Stomata are pores that let gases in and out of leaves. A leaf has 300 stomata per square millimetre. A small leaf measures 20 square millimetres in the studied patch. If 5% of the stomata are blocked by dust, how many stomata are still working in that patch? [Respiration in Organisms]

- **(A)** 5700
- **(B)** 5715
- **(C)** 4500
- **(D)** 285
- **(E)** 570
- **(F)** 300

70. Find 0.025 x 1000 - 12.5 / 0.5. [A Peek Beyond the Point]

- **(A)** 0
- **(B)** -12.5
- **(C)** -25
- **(D)** 50
- **(E)** 2.5
- **(F)** 12.5

71. The maximum temperature for a week totals 217 degrees Celsius across 7 days. If the hottest day was 35 degrees Celsius, what is the average of the OTHER 6 days? [Weather, Climate & Adaptations]

- **(A)** 36.333 deg C
- **(B)** 182 deg C
- **(C)** 30.333 deg C
- **(D)** 29 deg C
- **(E)** 31 deg C
- **(F)** 30 deg C

72. A computer file is 2^10 kilobytes. Given that 2^10 = 1024, what is the file size in kilobytes? [Exponents & Powers]

- **(A)** 210 KB
- **(B)** 512 KB
- **(C)** 20 KB
- **(D)** 100 KB
- **(E)** 1000 KB
- **(F)** 1024 KB
- **(G)** 2048 KB

73. The chest cavity changes size as we breathe. Suppose at rest the lungs hold 2.4 litres of air and a deep breath adds another 1.8 litres. By what percentage does the air in the lungs increase during that deep breath? [Respiration in Organisms]

- **(A)** 133.333%
- **(B)** 42.857%
- **(C)** 57.142%
- **(D)** 25%
- **(E)** 1.8%
- **(F)** 75%
- **(G)** 180%

74. A jug holds 2.4 litres. It is two-thirds full of water. How many 0.2-litre cups can be completely filled from the water in the jug? [A Peek Beyond the Point]

- **(A)** 16 cups
- **(B)** 4 cups
- **(C)** 10 cups
- **(D)** 8 cups
- **(E)** 6 cups
- **(F)** 12 cups
- **(G)** 9 cups

75. A region gets 1200 mm of rain a year. 65% of it falls in the 4 monsoon months. What is the AVERAGE monthly rainfall during those monsoon months? [Weather, Climate & Adaptations]

- **(A)** 780 mm
- **(B)** 195 mm
- **(C)** 260 mm
- **(D)** 420 mm
- **(E)** 65 mm
- **(F)** 300 mm

76. Find the value of (4^3 x 4^2) / 4^4. [Exponents & Powers]

- **(A)** 16
- **(B)** 4
- **(C)** 1
- **(D)** 8
- **(E)** 64
- **(F)** 4^5

77. Anaerobic respiration in our muscles makes lactic acid, but in yeast it makes alcohol and carbon dioxide. A juice is left to ferment and 46 grams of sugar give 23 grams of alcohol plus some carbon dioxide. What fraction of the sugar's mass became alcohol? [Respiration in Organisms]

- **(A)** 3/2
- **(B)** 1/3
- **(C)** 2/1
- **(D)** 2/3
- **(E)** 1/23
- **(F)** 23/69
- **(G)** 1/2

78. The sum of two decimals is 5.6 and one of them is 1.85. The smaller of the two is then multiplied by 0.4. What is this product? [A Peek Beyond the Point]

- **(A)** 2.24
- **(B)** 0.74
- **(C)** 0.5
- **(D)** 1.85
- **(E)** 0.46
- **(F)** 1.5

79. Monkeys in rainforests have long tails for balance and strong limbs to swing between trees. A monkey swings 2.5 metres per jump and makes 3 jumps then rests. If it must travel 45 metres, how many full sets of '3 jumps' does it complete? [Weather, Climate & Adaptations]

- **(A)** 4 sets
- **(B)** 18 sets
- **(C)** 9 sets
- **(D)** 5 sets
- **(E)** 7 sets
- **(F)** 45 sets
- **(G)** 6 sets

80. The mass of a dust grain is 7 x 10^-6 gram. Written as an ordinary decimal, this is how many grams? [Exponents & Powers]

- **(A)** 0.000007 g
- **(B)** 0.00007 g
- **(C)** 0.0007 g
- **(D)** 7000000 g
- **(E)** 0.0000007 g
- **(F)** 0.7 g
- **(G)** 0.000076 g

81. A diagram shows air passing nose, windpipe, then branching tubes to air sacs. If the windpipe splits into 2 main tubes, each of those into 2, and each of those again into 2, how many smallest tubes are there at the third level of branching? [Respiration in Organisms]

- **(A)** 3
- **(B)** 12
- **(C)** 6
- **(D)** 16
- **(E)** 8
- **(F)** 2
- **(G)** 4

82. A printer uses 0.012 litre of ink per page. A cartridge holds 0.3 litre. After printing some pages, 0.012 litre remains. How many pages were printed? [A Peek Beyond the Point]

- **(A)** 23 pages
- **(B)** 26 pages
- **(C)** 24 pages
- **(D)** 12 pages
- **(E)** 30 pages
- **(F)** 25 pages
- **(G)** 240 pages

83. A weather balloon rises and the temperature falls by 6 degrees Celsius for every 1 km of height. At the ground it is 30 degrees Celsius. What is the temperature at a height of 4.5 km? [Weather, Climate & Adaptations]

- **(A)** -3 deg C
- **(B)** 6 deg C
- **(C)** 24 deg C
- **(D)** 27 deg C
- **(E)** 0 deg C
- **(F)** 57 deg C
- **(G)** 3 deg C

84. Simplify 2^5 + 2^5 and write the answer as a single power of 2. [Exponents & Powers]

- **(A)** 4^5
- **(B)** 2^5
- **(C)** 2^25
- **(D)** 2^10
- **(E)** 2^7
- **(F)** 2^6

85. A patient's normal breathing rate is 15 per minute. A fever raises it by one-fifth, and a later infection raises that new rate by a further one-third. What is the breathing rate after both rises? [Respiration in Organisms]

- **(A)** 24
- **(B)** 23
- **(C)** 21
- **(D)** 20
- **(E)** 25
- **(F)** 18

86. Arrange in increasing order and pick the MIDDLE value: 0.305, 0.35, 0.035, 0.3, 0.503. [A Peek Beyond the Point]

- **(A)** 0.33
- **(B)** 0.35
- **(C)** 0.305
- **(D)** 0.3
- **(E)** 0.353
- **(F)** 0.035
- **(G)** 0.503

87. An animal's winter coat is 1.5 times as thick as its summer coat. If the summer coat is 8 mm thick, how much THICKER (in mm) is the winter coat than the summer coat? [Weather, Climate & Adaptations]

- **(A)** 16 mm
- **(B)** 6 mm
- **(C)** 1.5 mm
- **(D)** 20 mm
- **(E)** 12 mm
- **(F)** 4 mm
- **(G)** 3 mm

88. Evaluate (10^3 + 10^2) / 10^2. [Exponents & Powers]

- **(A)** 11
- **(B)** 110
- **(C)** 101
- **(D)** 10
- **(E)** 1000
- **(F)** 1.1
- **(G)** 100

89. Two students each blow into a balloon with one big breath. The first balloon holds 1.25 litres and the second holds 0.95 litre. If a healthy single breath out can be up to 1.5 litres, how much LESS than this maximum did the two students breathe out together? [Respiration in Organisms]

- **(A)** 1.2 litres
- **(B)** 0.3 litre
- **(C)** 3 litres
- **(D)** 0.55 litre
- **(E)** 0.2 litre
- **(F)** 0.8 litre

90. A shop weighs sugar in 0.25 kg packs. A sack of 40 kg is fully packed. Each pack is sold for Rs 22.5. What is the total money from selling all the packs? [A Peek Beyond the Point]

- **(A)** Rs 1800
- **(B)** Rs 3600 minus tax
- **(C)** Rs 360
- **(D)** Rs 7200
- **(E)** Rs 900
- **(F)** Rs 160
- **(G)** Rs 3600

91. In a city the chance of a sunny day is 0.6 and of a cloudy day is 0.25; the rest are rainy. Out of 40 days, how many rainy days would you expect? [Weather, Climate & Adaptations]

- **(A)** 4 days
- **(B)** 34 days
- **(C)** 10 days
- **(D)** 24 days
- **(E)** 15 days
- **(F)** 6 days

92. A chessboard puzzle puts 1 grain on the first square and doubles it each square. How many grains are on the 6th square? [Exponents & Powers]

- **(A)** 64
- **(B)** 6
- **(C)** 32
- **(D)** 12
- **(E)** 16
- **(F)** 36
- **(G)** 31

93. Respiration releases energy slowly inside cells, unlike a candle that burns fast. A sugar sample holds 200 energy units. A cell usefully captures 40% of them, while burning the same sugar usefully captures only 15%. How many MORE useful energy units does the cell capture than burning? [Respiration in Organisms]

- **(A)** 65 units
- **(B)** 80 units
- **(C)** 50 units
- **(D)** 20 units
- **(E)** 30 units
- **(F)** 25 units

94. A thermometer reads 36.8 in the morning and rises by 0.45 each hour for 4 hours, then falls by 0.7. What is the final reading? [A Peek Beyond the Point]

- **(A)** 37.9
- **(B)** 39.3
- **(C)** 38.0
- **(D)** 36.55
- **(E)** 37.2
- **(F)** 38.45
- **(G)** 38.6

95. The lowest night temperature in a hill town for 4 nights is -2, 1, -3 and 0 degrees Celsius. What is the average of these four readings? [Weather, Climate & Adaptations]

- **(A)** -1.5 deg C
- **(B)** -1 deg C
- **(C)** -4 deg C
- **(D)** 0 deg C
- **(E)** -6 deg C
- **(F)** -2 deg C

96. Simplify (a^3 x a^2) / a^4 where a is not zero, then find its value when a = 5. [Exponents & Powers]

- **(A)** 5^9
- **(B)** 625
- **(C)** 10
- **(D)** 125
- **(E)** 25
- **(F)** 1
- **(G)** 5

97. A spirometer shows a person breathes out 7.5 litres of air over 15 breaths. If carbon dioxide is 4% of exhaled air, how many millilitres of carbon dioxide are breathed out in total (1 litre = 1000 mL)? [Respiration in Organisms]

- **(A)** 3000 mL
- **(B)** 30 mL
- **(C)** 450 mL
- **(D)** 300 mL
- **(E)** 0.3 mL
- **(F)** 75 mL
- **(G)** 120 mL

98. A piece of wire is 9.6 metres long. From it, 1.25 metres is bent into a hook and the rest is cut into 5 equal straight pieces. How long is each straight piece? [A Peek Beyond the Point]

- **(A)** 1.6 m
- **(B)** 1.92 m
- **(C)** 8.35 m
- **(D)** 2.17 m
- **(E)** 1.67 m
- **(F)** 1.7 m
- **(G)** 1.45 m

99. A fish in a cold mountain stream stays active when its body uses 1.2 mL of oxygen per minute. In warmer water its use rises by one-fourth. How much oxygen does it use in 8 minutes in the warmer water? [Weather, Climate & Adaptations]

- **(A)** 12 mL
- **(B)** 1.5 mL
- **(C)** 2.4 mL
- **(D)** 15 mL
- **(E)** 10.8 mL
- **(F)** 24 mL

100. Simplify 3^2 x 3^2 x 3^2 and write it as a single power of 3, then give its value. [Exponents & Powers]

- **(A)** 3^8 = 6561
- **(B)** 3^6 = 729
- **(C)** 3^6 = 18
- **(D)** 27^2 only
- **(E)** 3^5 = 243
- **(F)** 3^4 = 81
- **(G)** 9^3 = 729 as power of 9